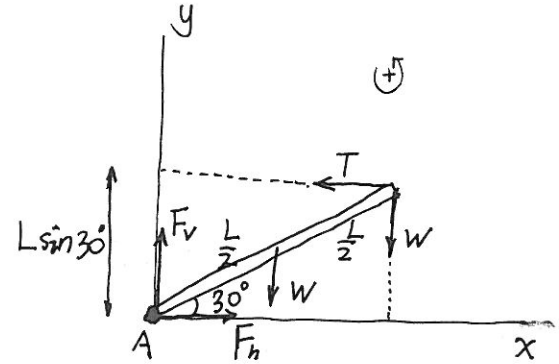
Phys. 2110 Solutions HW11 11.13, 14, 19, 26, 41, 42, 49, 59

11.13 (a) $\Sigma F_y = ma_y$

$F_v - W - W = 0 \qquad F_v = 2W$



Sum torques about point A.
The pivot force has zero moment arm for this axis and so doesn't enter into the torque equation.

$\tau_A = 0$

$TL\sin 30.0° - W\left(\left(\frac{L}{2}\right)\cos 30.0°\right) - W(L\cos 30.0°) = 0$

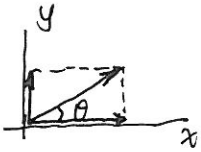
$T = \dfrac{3W\cos 30.0°}{2\sin 30.0°} = 2.60W$

Then $\Sigma F_x = ma_x$ implies $T - F_h = 0$ and $F_h = 2.60W$

Then we can find the magnitude and direction of $\vec{F}$

$F = \sqrt{F_h^2 + F_v^2} = \sqrt{(2.60W)^2 + (2.00W)^2} = 3.28W$

$\tan\theta = \dfrac{F_v}{F_h} = \dfrac{2.00W}{2.60W} \qquad \theta = 37.6°$

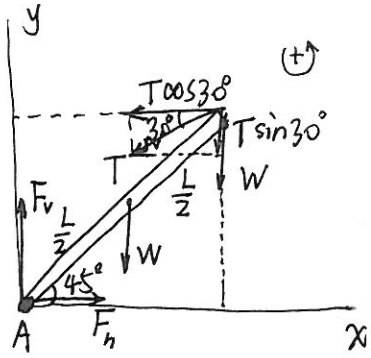


(b) $\Sigma \tau_A = 0 + (T\cos 30.0°)(L\sin 45.0°)$

$- (T\sin 30.0°)(L\cos 45.0°) - W\left(\frac{L}{2}\cos 45.0°\right)$

$- W(L\cos 45.0°) = 0$

$T = \dfrac{3W}{2(\cos 30.0° - \sin 30.0°)} = 4.10W$



$\Sigma F_x = ma_x$

$F_h - T\cos 30.0° = 0 \qquad F_h = T\cos 30.0° = 3.55W$

$\Sigma F_y = ma_y$

$F_v - W - W - T\sin 30.0° = 0$

$F_v = 2W + (4.10W)\sin 30.0° = 4.05W$

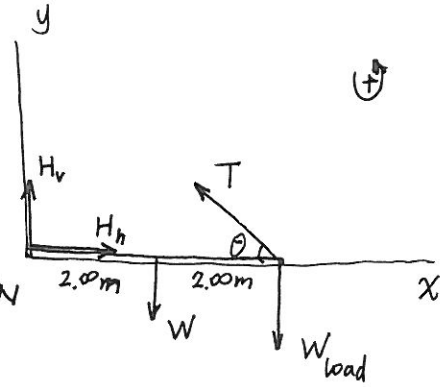
$F = \sqrt{F_h^2 + F_v^2} = 5.39W$

$\tan\theta = \dfrac{F_v}{F_h} = \dfrac{4.05W}{3.55W} \qquad \theta = 48.8°$

11.14 (a) $\Sigma \tau = 0$ gives

$$-W(2.00\,m) - W_{load}(4.00\,m) + T\sin\theta\,(4.00\,m) = 0$$

and $$T = \frac{(150\,N)(2.00\,m) + (300\,N)(4.00\,m)}{(4.00\,m)(0.600)} = 625\,N$$

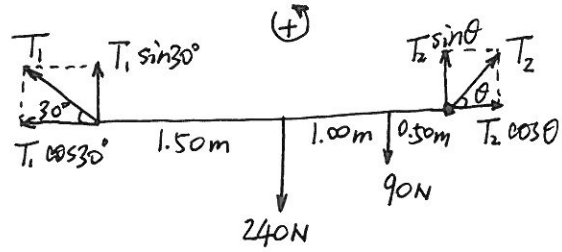


(b) $\Sigma F_x = 0$ gives $H_h = T\cos\theta = (625\,N)(0.800) = 500\,N$

$\Sigma F_y = 0$ gives $H_v = W + W_{load} - T\sin\theta$

$$= 150\,N + 300\,N - (625\,N)(0.600) = 75\,N$$

11.19 $\Sigma \tau_z = 0$, axis at right end of rod



$$(240\,N)(1.50\,m) + (90\,N)(0.50\,m) - (T_1 \sin 30°)(3.00\,m) = 0$$

$$T_1 = \frac{360\,N\cdot m + 45\,N\cdot m}{1.50\,m} = 270\,N$$

$\Sigma F_x = m a_x$

$T_2 \cos\theta - T_1 \cos 30° = 0$ and $T_2 \cos\theta = T_1 \cos 30° = 234\,N$

$\Sigma F_y = m a_y$

$T_1 \sin 30° + T_2 \sin\theta - 240\,N - 90\,N = 0$

$T_2 \sin\theta = 330\,N - (270\,N)\sin 30° = 195\,N$

Then $\dfrac{T_2 \sin\theta}{T_2 \cos\theta} = \dfrac{195\,N}{234\,N}$ $\quad \tan\theta = 0.8333$ and $\theta = 40°$

$$T_2 = \frac{195\,N}{\sin 40°} = 303\,N$$

11.26 $$Y = \frac{l_0 F_\perp}{A\,\Delta l}$$

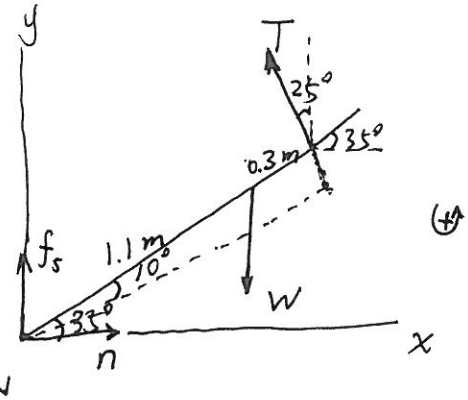
$A = \pi r^2 = \pi(3.5\times10^{-3}\,m)^2 = 3.85\times10^{-5}\,m^2$

$F_\perp = (65.0\,kg)(9.80\,m/s^2) = 637\,N$

$$Y = \frac{(45.0\,m)(637\,N)}{(3.85\times10^{-5}\,m^2)(1.10\,m)} = 6.77\times10^{8}\,Pa$$

11.41 (a) $\Sigma \tau_z = 0$ gives $T(1.4\,m)\cos 10^\circ - W(1.1\,m)\cos 35.0^\circ = 0$

$$T = \frac{(1.1\,m)\cos 35.0^\circ}{(1.4\,m)\cos 10^\circ}(82.0\,kg)(9.80\,m/s^2) = 525\,N$$



(b) $\Sigma F_x = 0$ gives $n = T\sin 25.0^\circ = 222\,N$

$\Sigma F_y = 0$ gives $f_s + T\cos 25^\circ - W = 0$

$f_s = (82.0\,kg)(9.80\,m/s^2) - (525\,N)\cos 25^\circ = 328\,N$

(c) $\mu_s = \frac{f_s}{n} = 1.48$

11.42 If Lancelot were at the end of the bridge, the tension in the cable T would be $T(12.0\,m) = (600\,kg)(9.80\,m/s^2)(12.0\,m) + (200\,kg)(9.80\,m/s^2)(6.0\,m)$

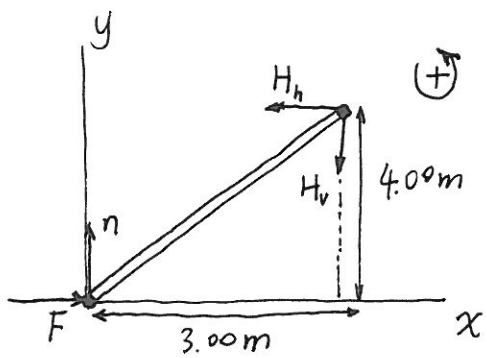
$T = 6860\,N > 5.80\times10^3\,N$, so the cable will break

To find the distance x Lancelot can ride

$(600\,kg)(9.80\,m/s^2)\,x = (5.80\times10^3\,N)(12.0\,m) - (200\,kg)(9.80\,m/s^2)(6.0\,m)$

$x = 9.84\,m$

11.49



$\Sigma F_x = ma_x$

$F = H_h = 120\,N$

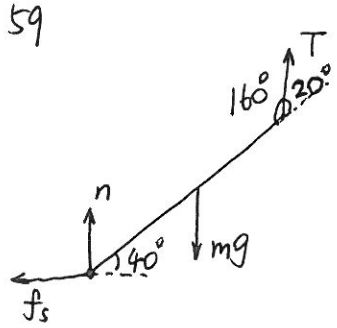
$\Sigma F_y = ma_y$

$n - H_v = 0$

$\Sigma \tau = 0$ gives $F(4.00\,m) - n(3.00\,m) = 0$

$n = (4.00\,m/3.00\,m)(120\,N) = 160\,N$ and then $H_v = 160\,N$

11.59



(a) $\Sigma \tau = 0$, axis at lower end of beam

Let the length of the beam be L

$T(\sin 20^\circ)L - \frac{L}{2}mg\cos 40^\circ = 0$

$T = 2740\,N$

(b) $\Sigma F_y = 0$ gives $n - W + T\sin 60^\circ = 0$ so $n = 73.6\,N$

$\Sigma F_x = 0$ gives $f_s = T\cos 60^\circ = 1372\,N$

$\mu_s = \frac{f_s}{n} = 19$